

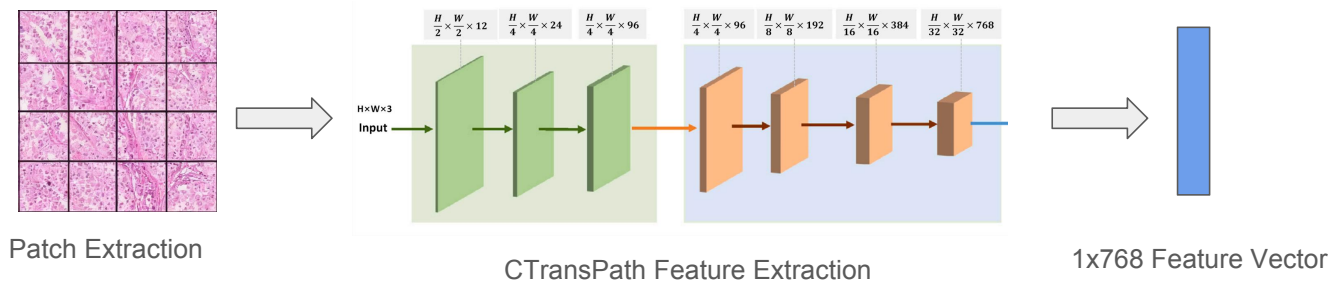


Presentor: Lillian Lam

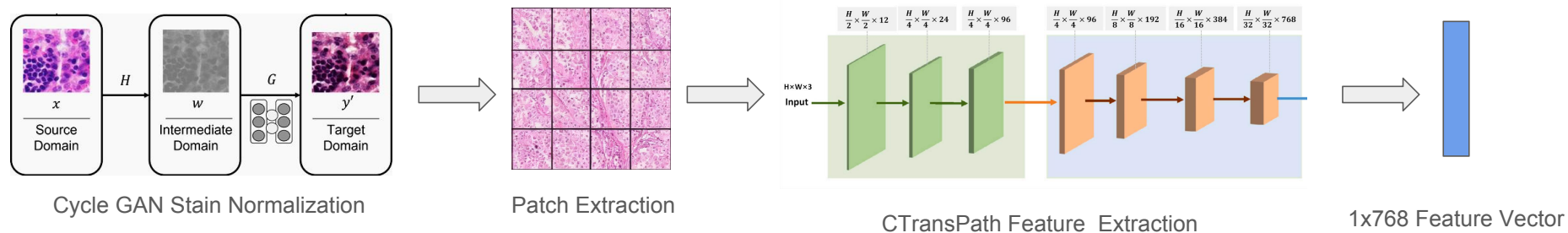
Members: Yang Cao, Naina Misra, Chaohui (Polaris) Xu

Recap

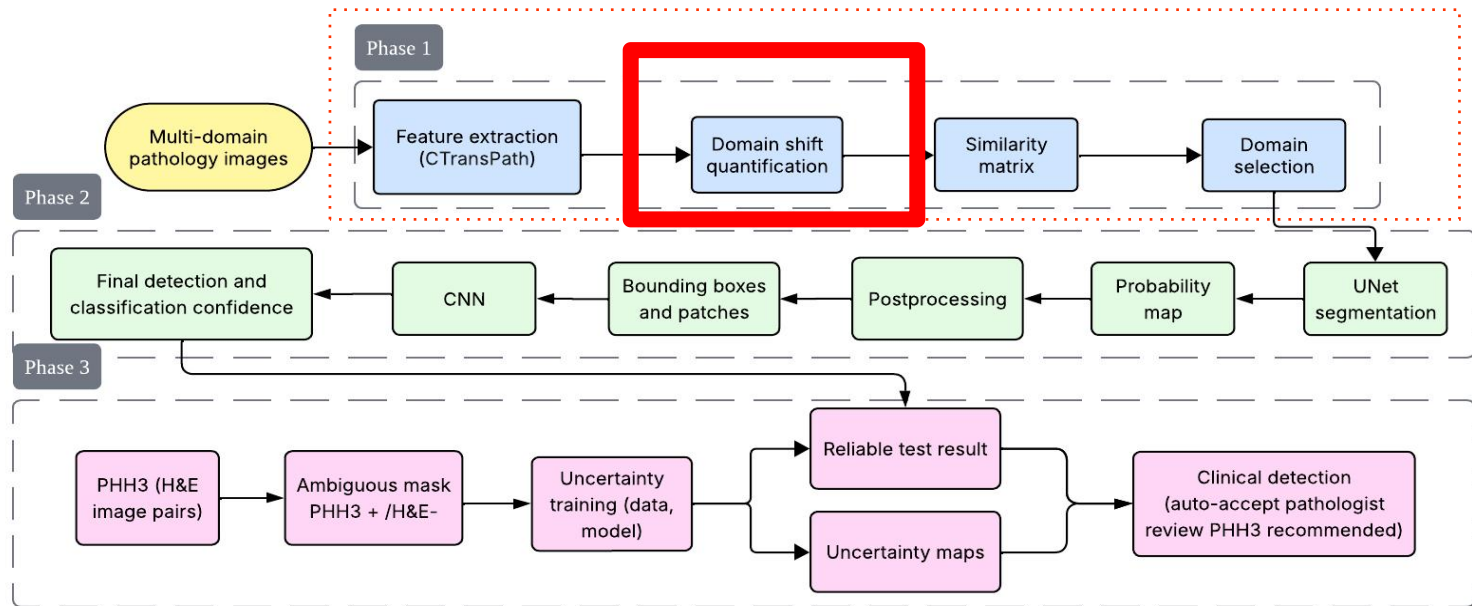
CTransPath Feature Extraction



Stain Normalized CTransPath Feature Extraction



Pipeline



Maximum Mean Discrepancy (MMD)



Set X: 100 images of NYC



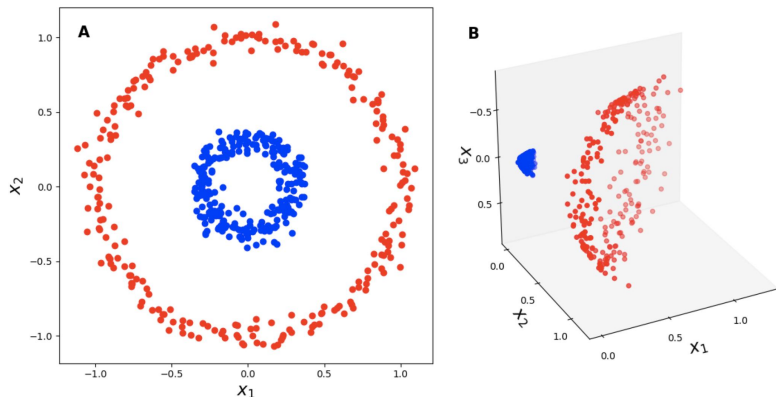
Set Y: 100 images of HK

MMD measures: How different are these collections on average?

$$\text{MMD}^2 = \frac{1}{n(n-1)} \sum_i \sum_{j \neq i} k(x_i, x_j) + \frac{1}{m(m-1)} \sum_i \sum_{j \neq i} k(y_i, y_j) - \frac{2}{nm} \sum_i \sum_j k(x_i, y_j)$$

Kernel Trick

MMD uses a kernel function to compare distributions in a high-dimensional space where differences are easier to see.



Kernel Choice: Linear Kernel (basic dot product)

$$k(x, y) = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

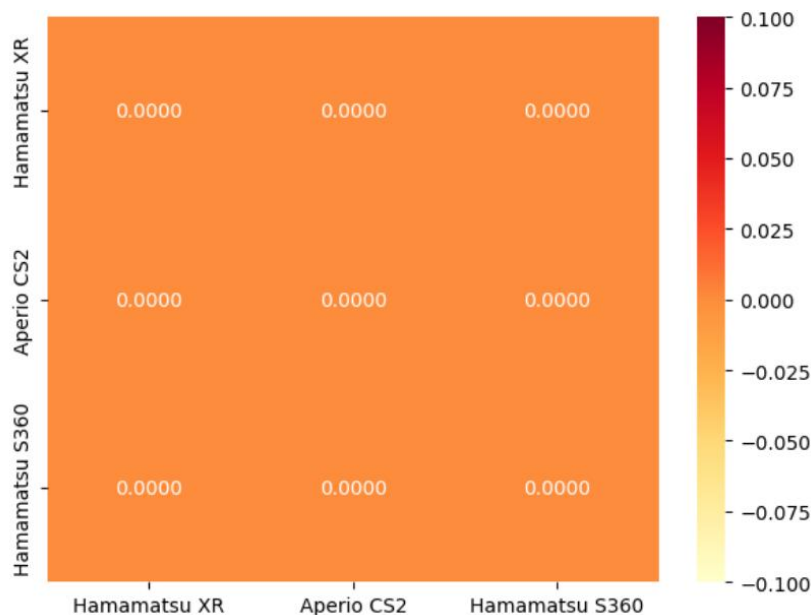
- CTransPath features already high-dimensional (768D)
- Computationally efficient - no hyperparameters (gaussian kernel requires γ)
- Interpretable - measures similarity in original space
- Stable - no gamma parameter tuning needed

Problem: Features have small means

Scanner Hamamatsu XR:
Samples: (29952, 768)
Feature mean: 0.0005
Feature std: 0.1338
Any NaN: False

Scanner Aperio CS2:
Samples: (23142, 768)
Feature mean: 0.0008
Feature std: 0.1341
Any NaN: False

Scanner Hamamatsu S360:
Samples: (28015, 768)
Feature mean: 0.0005
Feature std: 0.1380
Any NaN: False



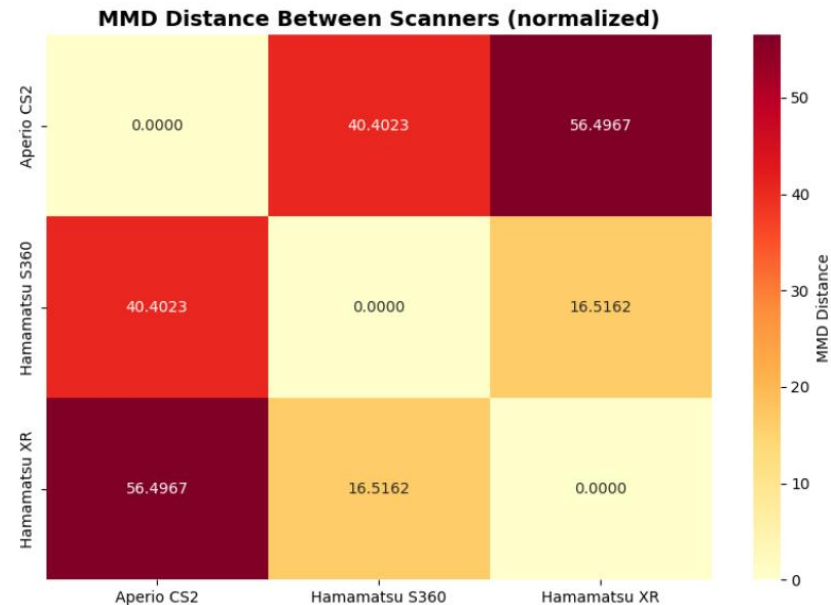
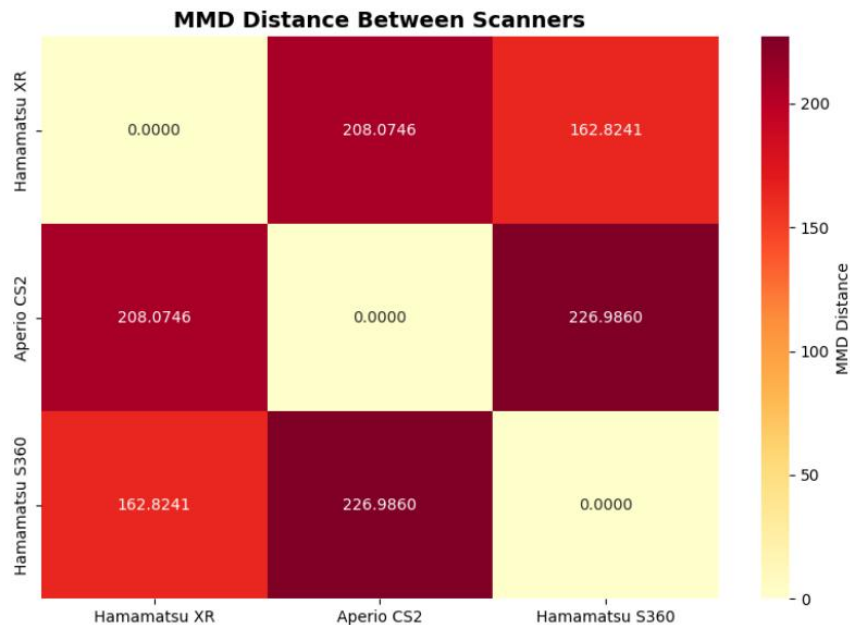
Dot product of small number $\Rightarrow$ small results (floating-point precision limits)

Solution: Feature Scaling (divide features by pooled standard deviation)

Scanner Differences

Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
1a	50	breast carcinoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$
1b	50	breast carcinoma	UMC Utrecht	human	Hamamatsu S360 (0.5 N/A)	$0.23 \frac{\mu m}{px}$
1c	50	breast carcinoma	UMC Utrecht	human	Leica ScanScope CS2	$0.25 \frac{\mu m}{px}$

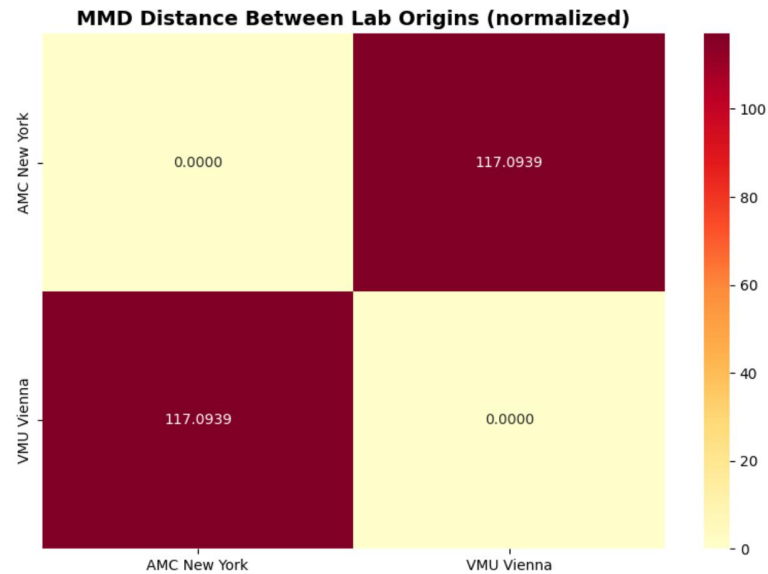
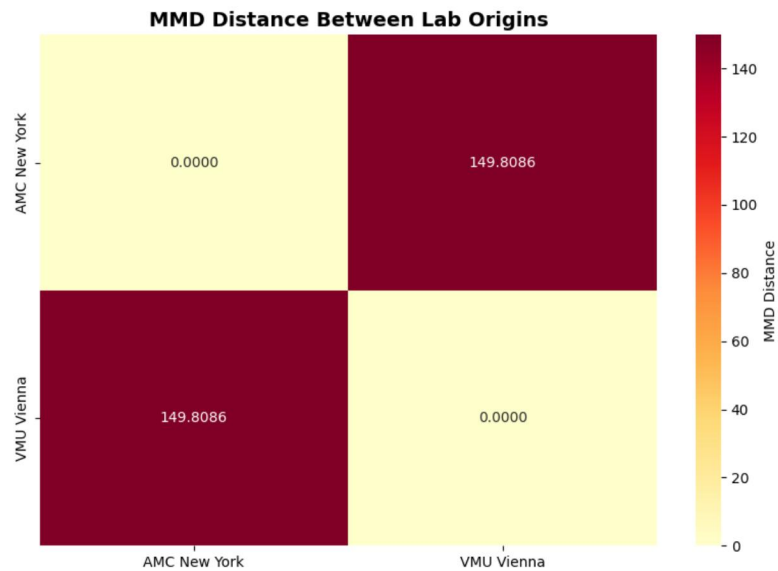
Scanner Differences



Lab Origin

Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
6a	85	soft tissue sarcoma	AMC New York	canine	3DHistech Pannoramic Scan II	$0.25 \frac{\mu m}{px}$
6b	15	soft tissue sarcoma	VMU Vienna	canine	3DHistech Pannoramic Scan II	$0.25 \frac{\mu m}{px}$

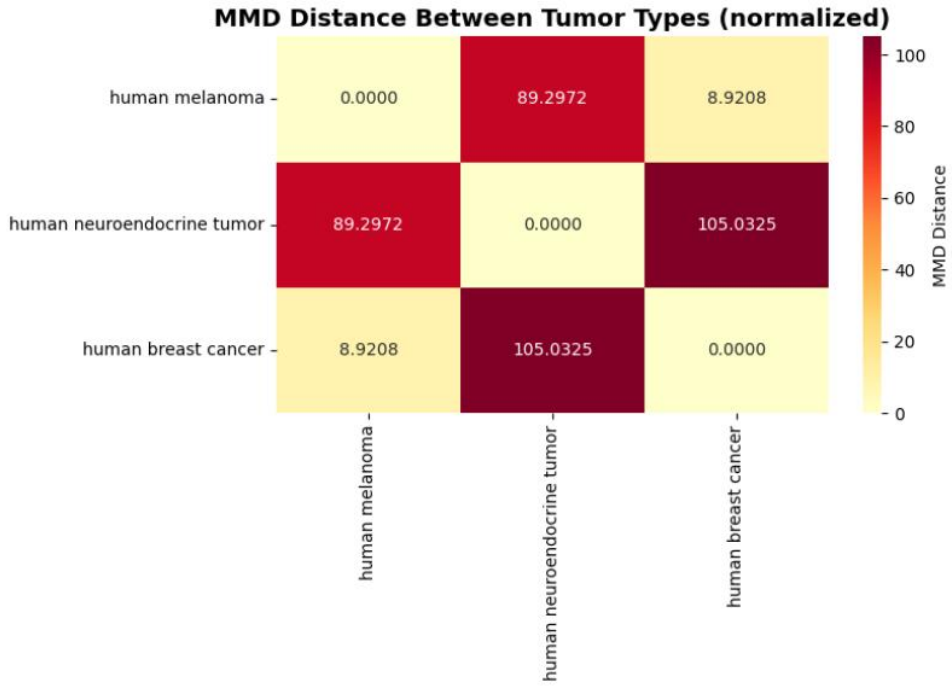
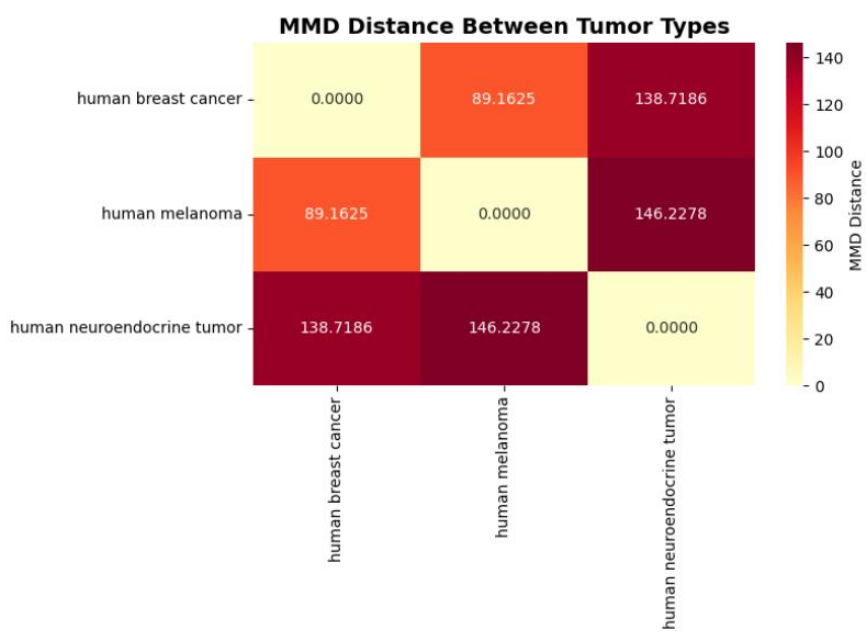
Lab Origin



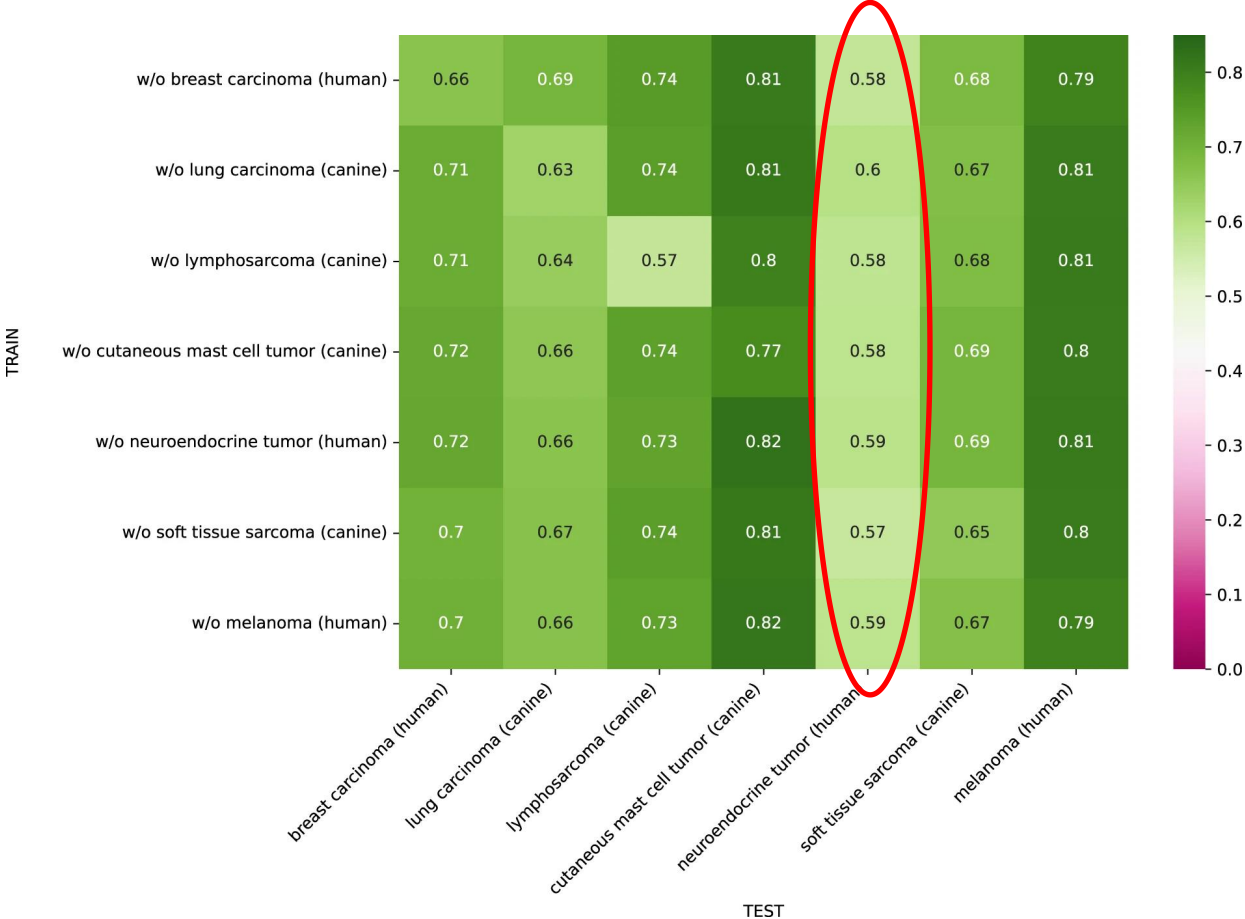
Tumor Differences

Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
1a	50	breast carcinoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$
5	55	neuroendocrine tumor	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$
7	49	melanoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$

Tumor Differences



Leave-One-Domain-Out Training from Paper



Summary

- Normalization reduced the MMD in similar Scanner and Tumor Type
- Slightly reduced MMD within lab origin

Next Steps

- Domain shift similarity with CORAL
- Clustering within feature space (look at Silhouette Score)

Thank
You